

Keon E

613-793-1733 | ke@uwaterloo.ca | [linkedin.com/in/keon-e](https://www.linkedin.com/in/keon-e) | github.com/keon-e

SUMMARY OF QUALIFICATIONS

- Developed and implemented AI/ML models from scratch in C++, training a CNN to **98.02%** validation accuracy on MNIST without any ML framework
- Strong programming skills in C++ and Python
- Basic understanding of data preprocessing and model evaluation, built by implementing training pipelines by hand
- Quick to learn and apply new technologies, independently learning cluster infrastructure as a WATcloud contributor
- Works well independently and as part of a team, with experience supervising staff

EDUCATION

University of Waterloo

Waterloo, ON

Honours Mathematics, Co-operative Program

Aug. 2026 – Expected 2031

Relevant Coursework: Designing Functional Programs (CS 135), Algebra (Advanced) (MATH 145), Calculus 1 (Advanced) (MATH 147)

Euclid Mathematics Contest: Distinction, top 25% (2026)

PROJECTS

Tensor Autograd Engine | C++

2026

- Built a tensor autograd engine from scratch, extending a scalar micrograd-style implementation to N-dimensional tensors with broadcasting, matrix multiplication, convolution, and pooling
- Implemented a full training pipeline and CNN on top of the engine with no external ML framework
- Trained and evaluated an MLP to **97.34%** and a CNN to **98.02%** validation accuracy on MNIST
- Implemented backpropagation and tensor memory layout by hand, rather than relying on frameworks

EXPERIENCE

WATcloud

Waterloo, ON

Team Member, Student-Run Compute Cluster

Sep. 2026 – Present

- Learned cluster infrastructure and internal documentation independently in a student-run team

City of Ottawa

Ottawa, ON

Wading Pool Supervisor

Summer 2025

- Promoted to Supervisor after two summers as Pool Attendant
- Supervised attendant staff and oversaw daily pool operations, enforcing safety protocols and City guidelines

Pool Attendant

Summers 2023 – 2024

- Monitored swimmer safety and supported site staff in daily wading pool operations

TECHNICAL SKILLS

Languages: C++, C, Python, Racket

Machine Learning: Backpropagation, neural networks (MLP, CNN), data preprocessing, model evaluation

Tools: Git, g++

EXTRACURRICULAR ACTIVITIES

Lisgar Collegiate Institute: Varsity Basketball (2025 – 2026), Track and Field (2023 – 2026), Symphonic Winds, Bb Clarinet (2022 – 2026)